

An Efficient Approach for Cell Tower Performance Measurement Using Autonomous Aerial Vehicle

Pritam Prayash Behra, Faiza Arfin, Somen Senapati, Sipra Swain and Rakesh Ranjan Swain

Department of Computer Science and Engineering

Institute of Technical Education and Research (ITER),

Siksha 'O' Anusandhan (Deemed to be University), Bhubaneswar, Odisha, India

Abstract—In an era where automation and cost efficiency are held on a pedestal, unmanned aerial vehicles (UAVs) are becoming a practical tool for high-resolution measurement of cellular network performance. They enable flexible, low-cost data collection across complex geographic regions. Recent UAV measurement studies demonstrate that Reference Signal Received Power (RSRP) and Reference Signal Received Quality (RSRQ) exhibit strong spatial structure and when measurement redundancy is reduced using spatial aggregation, can be modeled effectively. However, dense UAV datasets can create noisy measurements, irrelevant spatial correlation, and high computational cost for point-wise modeling. To address these issues, we propose a bin-based quantization of latitude and longitude to form grids across a region, extract statistical features per grid, and compute correlations to reveal inter-grid dependencies. The master grids are identified and used to predict the slave grids' signal, before applying Travelling Salesman Problem (TSP) and Ant Colony Optimization (ACO) to minimize distance travelled and energy costs.

I. INTRODUCTION

In an era where automation and cost efficiency are held on a pedestal, UAVs are becoming a practical tool for high-resolution measurement of cellular network performance. They enable flexible and low-cost data collection across complex geographic regions.

II. LITERATURE SURVEY

III. PROPOSED WORK

A. Proposed Framework

This work proposed an approach for predicting slave grids' RSRP applying the correlation features collected using UAV.

B. Data Collection

Data was collected via roadway measurement including latitude, longitude and radio indicators such as RSRP, RSRQ, RSSI, SNR and SINR.

C. Grid Formation

The target area is divided into fixed grids using latitude and longitude bin-based quantization.

D. Master-Slave Grid Selection

E. Feature Engineering

Haversine distance and nonlinear transformations were used. Hybrid feature selection combining permutation importance, RFE and embedded methods was applied.

Algorithm 1 Master-Slave Grid Selection

- 1: Load correlation matrix C of active nodes A_n
 - 2: Set correlation threshold τ
 - 3: **while** $A_n \neq \emptyset$ **do**
 - 4: Compute total correlation score: $S_i = \sum_j C_{ij} - C_{ii}$
 - 5: Select master: $G_m = \arg \max(S_i)$
 - 6: Identify slaves: $G_s = \{G_j \mid C_{mj} > \tau, j \neq m\}$
 - 7: Store mapping $M(G_m) = G_s$
 - 8: Remove G_m and G_s from A_n
 - 9: **end while**
 - 10: Output mapping M
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F. Model Development

Five models were evaluated: Extra Trees, XGBoost, SVR, Random Forest and CatBoost. Threshold $\tau = 0.7$ provided the best balance.

G. Path Planning and Energy Optimization Framework

Grid centroid:

$$C = \left(\frac{lat_{min} + lat_{max}}{2}, \frac{lon_{min} + lon_{max}}{2} \right)$$

Coordinate conversion:

$$x_i = (lon_i - lon_0) \times 111.32 \cos(lat_0)$$

$$y_i = (lat_i - lat_0) \times 110.574$$

Distance:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}$$

Algorithm 2 Path Planning

- 1: Load grid centroids V
 - 2: Convert lat-lon to planar coordinates
 - 3: Construct graph $G(V, E)$
 - 4: Compute distance d_{ij}
 - 5: Initialize $m = |V|$, $T = 1600$, $\alpha = 1$, $\beta = 2$, $\rho = 0.5$
 - 6: Initialize pheromone $\tau_{ij} = 1$
 - 7: $P^* \leftarrow ACO(G)$
 - 8: $D_{total} = \sum_{(i,j) \in P^*} d_{ij}$
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H. Energy Optimization

Distance energy:

$$E_{ij}^{dist} = P_{cruise} \frac{d_{ij}}{V} + C_{corr}$$

Deviation angle:

$$\begin{aligned}\vec{v}_{ij} &= (x_j - x_i, y_j - y_i, z_j - z_i) \\ \vec{v}_{jk} &= (x_k - x_j, y_k - y_j, z_k - z_j) \\ \theta_{ijk} &= \cos^{-1} \left(\frac{\vec{v}_{ij} \cdot \vec{v}_{jk}}{\|\vec{v}_{ij}\| \|\vec{v}_{jk}\|} \right)\end{aligned}$$

Turning energy:

$$E_{ijk}^{turn} = C_{turn} \frac{\theta_{ijk}}{\theta_{ref}}$$

Total transition energy:

$$E_{ijk} = E_{ij}^{dist} + E_{ijk}^{turn}$$

Algorithm 3 Energy Optimization

- 1: Perform Steps 1–4 of Path Planning
 - 2: Initialize UAV parameters: $V = 2$, $P_{cruise} = 308.709$,
 $C_{turn} = 225$, $\theta_{ref} = 2.1$, $C_{corr} = -0.852$
 - 3: Compute E_{ij}^{dist}
 - 4: Compute θ_{ijk}
 - 5: Compute E_{ijk}^{turn}
 - 6: Compute E_{ijk}
 - 7: Initialize $m = |V|$, $T = 1600$, $\alpha = 1$, $\beta = 2$, $\rho = 0.5$,
 $Q = 1$
 - 8: $\tau_{ij} = 1$
 - 9: Construct energy-weighted graph $w(i, j, k) = E_{ijk}$
 - 10: $P^* \leftarrow ACO(G)$
 - 11: $D_{total} = \sum_{(i,j) \in P^*} d_{ij}$
 - 12: $E_{total} = \sum_{(i,j,k) \in P^*} E_{ijk}$
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I. Normal TSP+ACO Energy Calculation

For path

$$P^* = \{n_0, n_1, \dots, n_m\}$$

$$E_{total} = \sum_{i=0}^{m-1} \left(P_{cruise} \frac{d_{i,i+1}}{V} + C_{corr} \right) + \sum_{i=1}^{m-1} \left(C_{turn} \frac{\theta_i}{\theta_{ref}} \right)$$

where $d_{i,i+1}$ is in meters and θ_i is in radians.

Parameters:

$$V = 2 \text{ m/s}, \quad P_{cruise} = 308.709 \text{ W}, \quad C_{turn} = 225, \quad \theta_{ref} = 2.1, \quad C_{corr} = -0.852$$